

Practice Problems 1

Not to turn in, but may be helpful before the exam

1. (adapted from book problems Chapter 3, #4 and #5, data from Example 3.4) The file `PP1 Problem1.R` contains code to construct a data set with 3835 observations, with one line in the data frame for each of the surveys returned. The 8th column, `femid`, is an identifier for whether the respondent was female (1) or male (0).
 - a. Add a 9th column with the sampling weight.
 - b. Using **only** columns 8 and 9, calculate $\hat{t}_{str}$ (i.e., don't use the **survey** package), the estimated total number of women.
 - c. Using **only** columns 8 and 9, calculate $\hat{p}_{str}$ (i.e., don't use the **survey** package), the estimated proportion of women.
 - d. Explain why it is not possible to calculate a confidence interval for p using only columns 8 and 9.
 - e. The additional information referenced in #5 has already been added to the dataframe. Using all available information, find a 95% confidence interval for the proportion of people who agree with the statement about rarely finding an article that interests them. (Hint: you may need to do some dataframe manipulation.)
2. Book problem Chapter 3, #9 (objects in bins). The R Companion to the textbook (pdf may be downloaded from Dr. Lohr's website—the link is in Canvas) gives instructions for using R to select a stratified random sample. Use the seed 724845.